

Network Analysis on Rumination

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2. Methods

2.1 Network analysis

Psychological networks have their own characteristics which differentiate them from other networks of the graph theory (Epskamp et al., 2017; Newman, 2010). Nodes are normally some measures or items of a psychological assessment tool, the edges of the network reflect their estimates of association derived from the data.

2.1.1 Removing redundant items As a preparation step, we applied the *goldbricker* function from the R package *networktools* to search for pairs of items in the RRS, SCRS and MHS items which were highly correlated ($r > 0.50$) and showed highly the same pattern with other items. When such “bad pairs” were identified, reduction techniques were put into use (with the *net_reduce* function in R): instead of the pair, the first principal component of the two variables were kept in the data as a new, merged variable.

The RRS items required no reduction steps as the highly correlating items did not extend their pattern to more nodes (their correlations to other nodes did not violate the criteria that $> 75\%$ of these correlations did not differ significantly).

2.1.2 Network Analysis with LASSO For the network analysis, the tuning parameter of *gamma* in the graphical LASSO algorithm has been set to 0.5 value in accordance with (Bernstein et al., 2019) in order to achieve sparse graphs. The final networks were chosen by the lowest values of the *extended Bayesian Information Criterion* (EBIC).

2.1.3. Community detection In order to get an understanding of community of nodes and to see if nodes can be grouped into neighbourhoods, we used the *igraph* R package. When detecting possible communities of nodes, the detection happened with the following parameters in the *spinglass algorithm*: TODO: try the walktrap algorithm from igraph!

2.1.4. Expected Influence of nodes To understand certain items’ centrality and importance, besides regularly used centrality measures (such as strength, closeness and betweenness of nodes in a network), one can also use the One- and Two-Step Expected Influence (EI) measures (Robinaugh et al, 2016). Traditionally in unweighted networks (where network edges are encoded to 0 or 1) a node’s centrality is the sum of edges, and in weighted nodes it is the sum of the edge weights. In this approach negative edges are treated the same way as positives, as the absolute values are used for centrality estimation. On the other hand, the Expected Influence metrics take into account if a node has positive and negative edges, too.

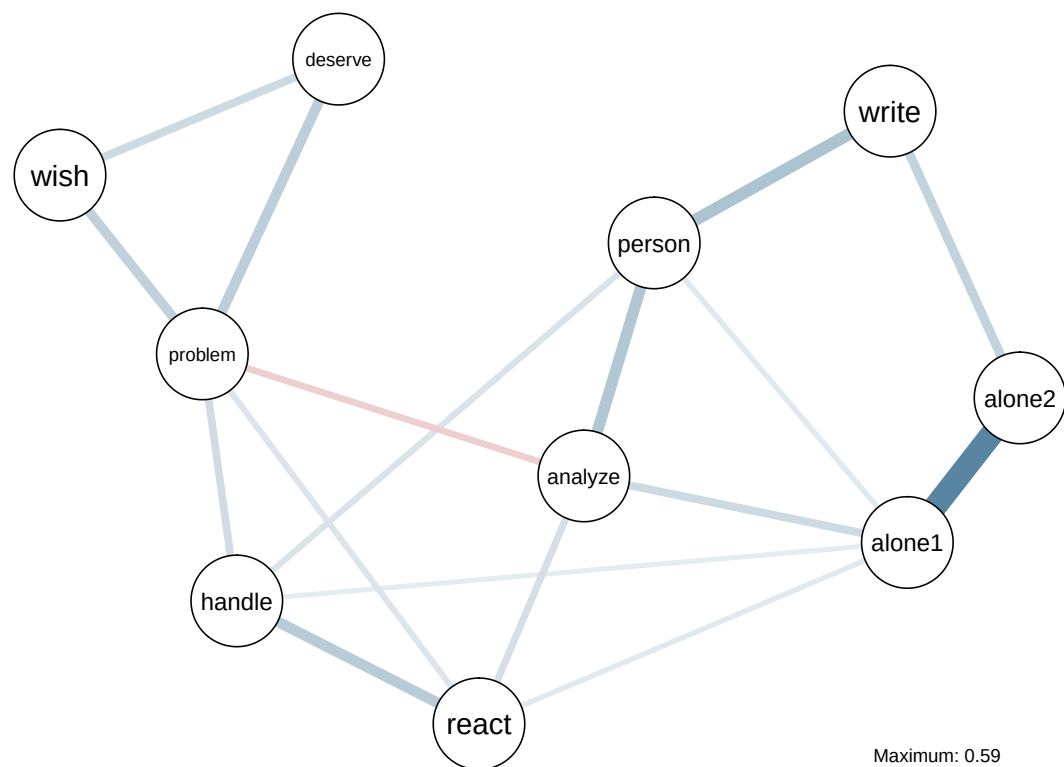
2.1.5. Identifying important nodes In order to further elaborate the question of importance within the network, we identified *bridge nodes* using the functions provided by the *networktools* R package (Jones, 2017). We estimated both one-step and two-step bridge expected influence measures which reflect the sum of edge weights connecting one node to other groups on nodes (or communities of nodes); and also express

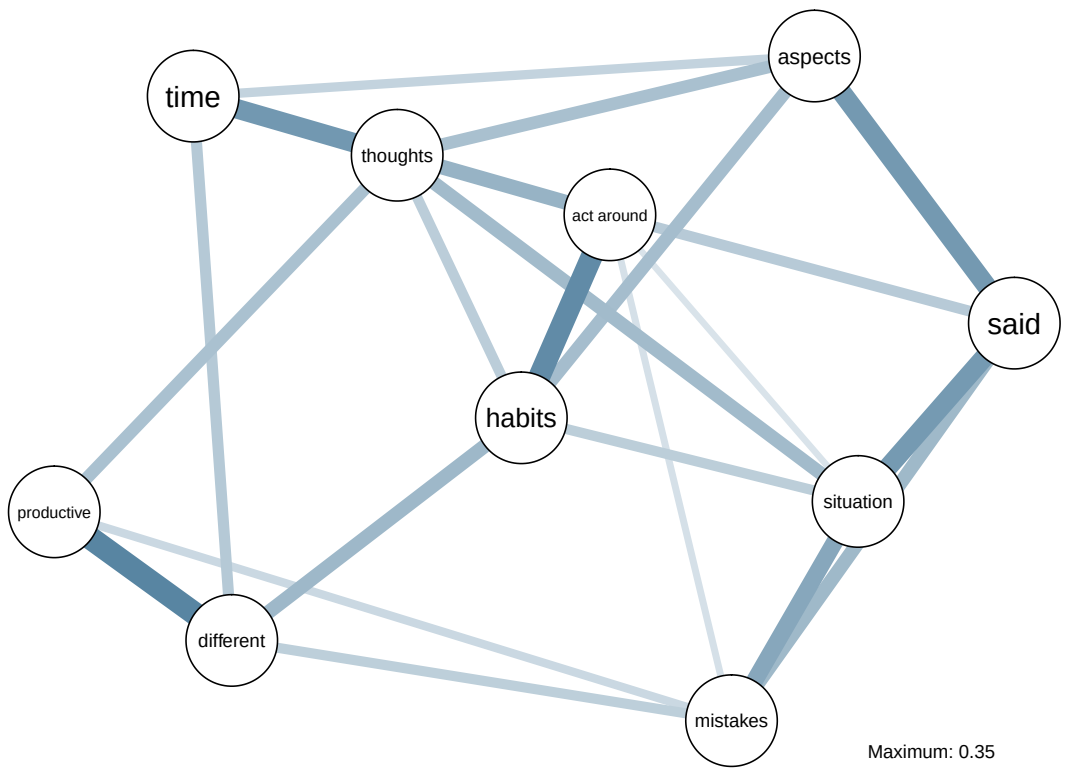
a node's secondary influence on the rest of the network, respectively. Higher influence scores indicate that a certain node is more likely to active neighbouring nodes, subgroups of nodes or communities.

3. Results

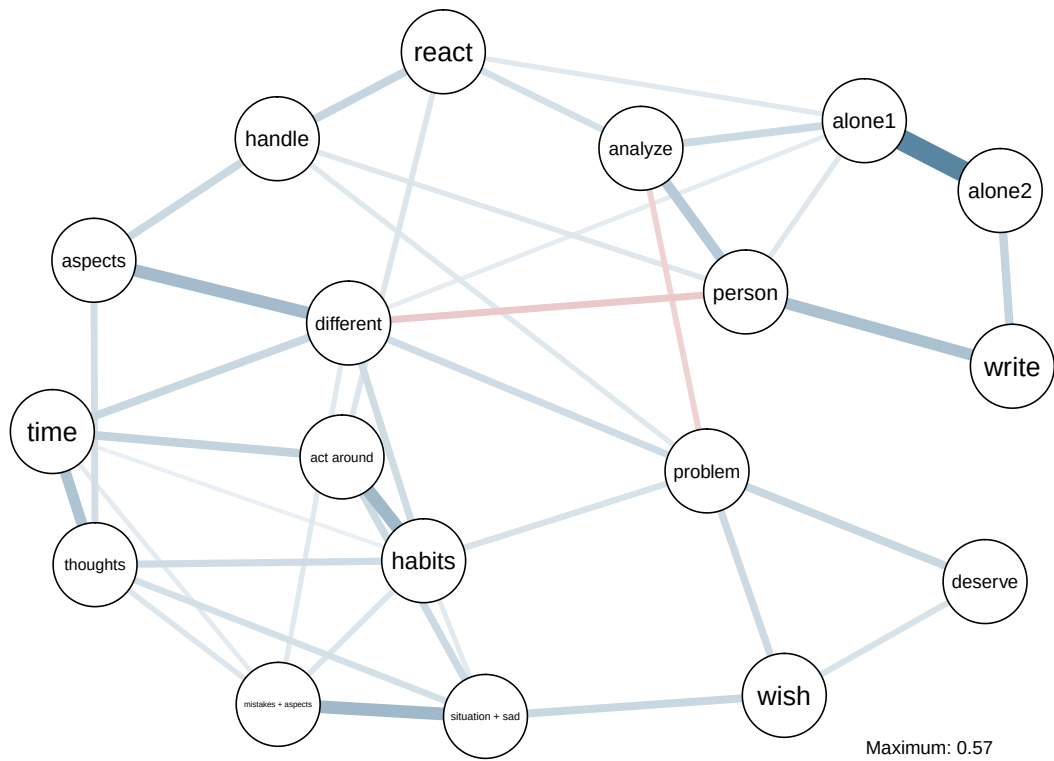
3.1 Network plots

RSS network





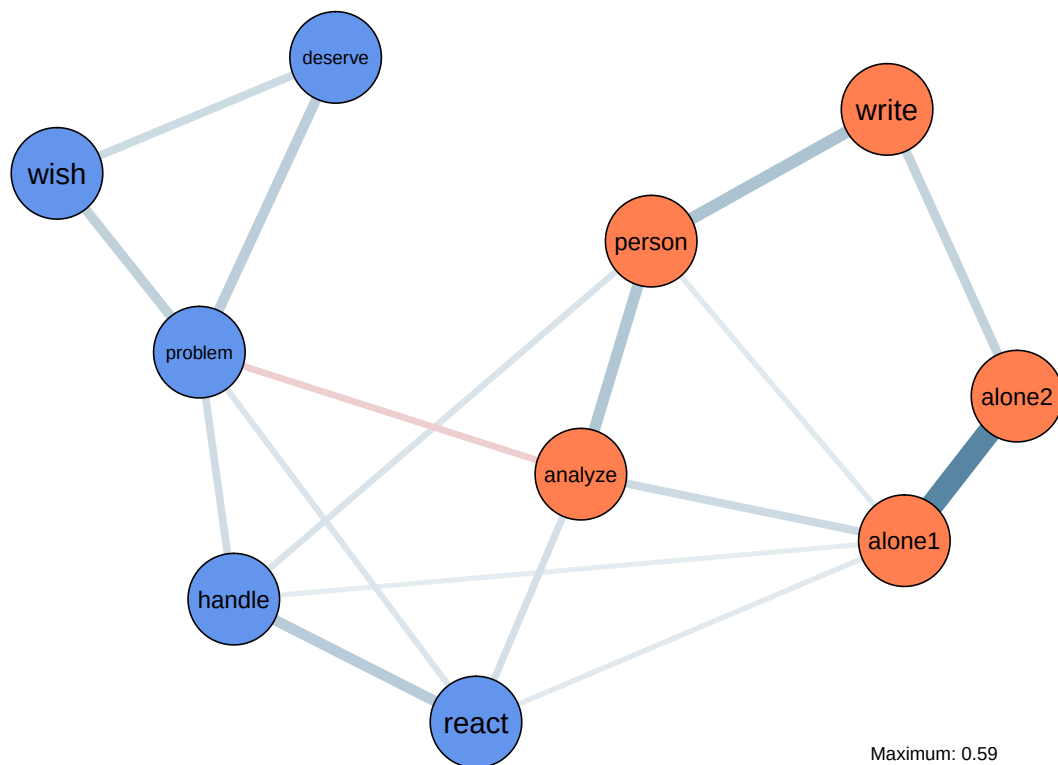
```
## |
## SCRS_4 & SCRS_2 SCRS_8 & SCRS_1
## 0.1666667 0.2222222
```



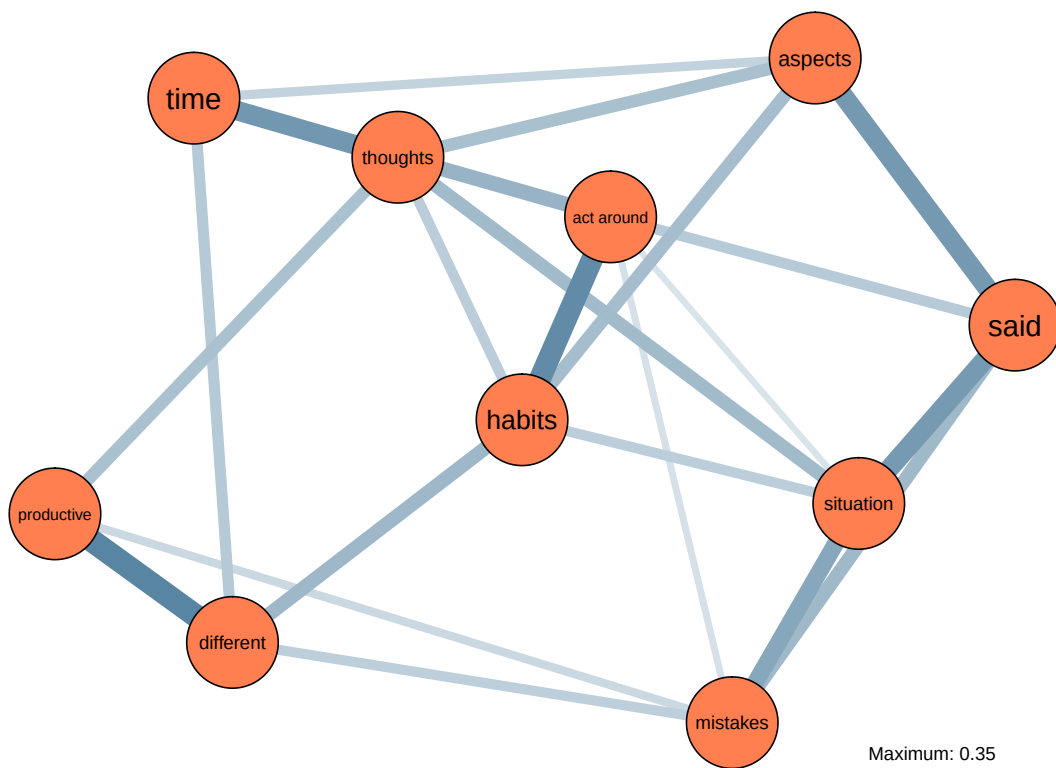
3.2 Community detection

Community mapping of RSS, MHC and SCRS with the *spinglass* algorithm:

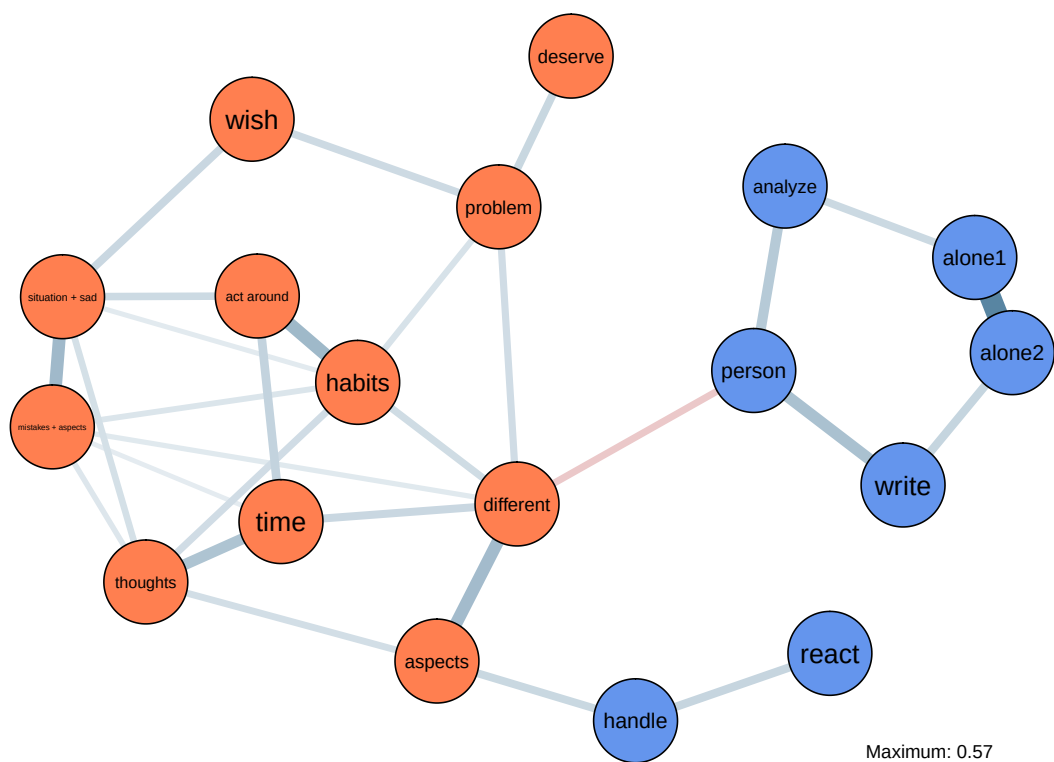
RSS communities identified with parameters $\gamma = 0.5$, $\text{spins} = 25$, $\text{starting temperature} = 1$, $\text{ending temperature} = 0.01$, $\text{cooling factor} = 0.99$.



SCRS communities identified with parameters $\gamma = 0.5$, $\text{spins} = 25$, $\text{starting temperature} = 1$, $\text{ending temperature} = 0.01$, $\text{cooling factor} = 0.99$.



RRS and MHC combined communities identified with parameters $\gamma = 0.5$, $\text{spins} = 25$, starting temperature = 1, ending temperature = 0.01, cooling factor = 0.99.



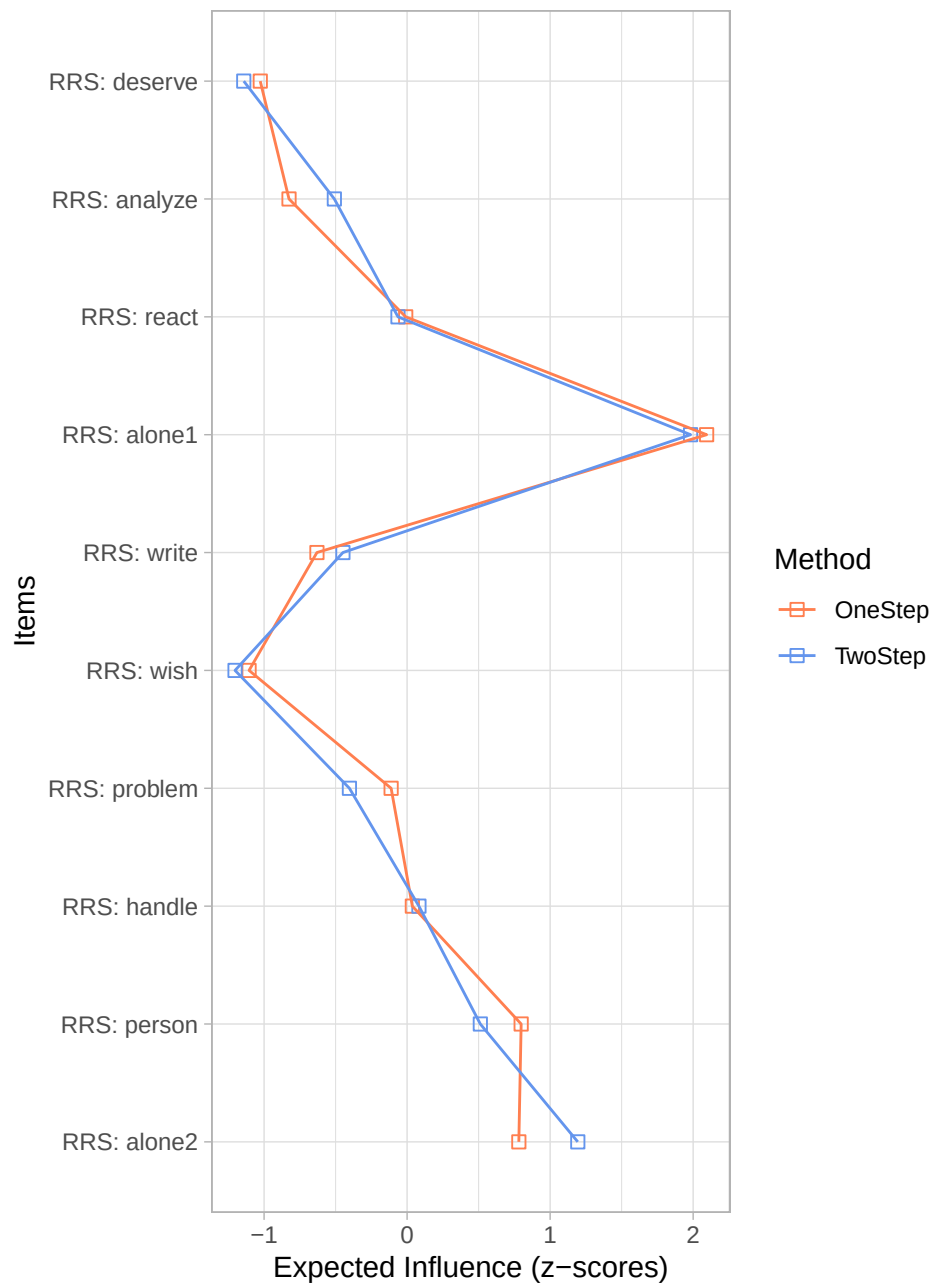
3.3 Expected Influence

In order to estimate the Expected Influence metrics, we used the *expectedInf* function of the *networktools* R package. The influence score are normalized.

RRS Expected Influence Measures

```
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
```

One- and Two-Step Expected Influence estimates for RR



SCRS Expected Influence Measures

```
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
```

One- and Two-Step Expected Influence estimates for

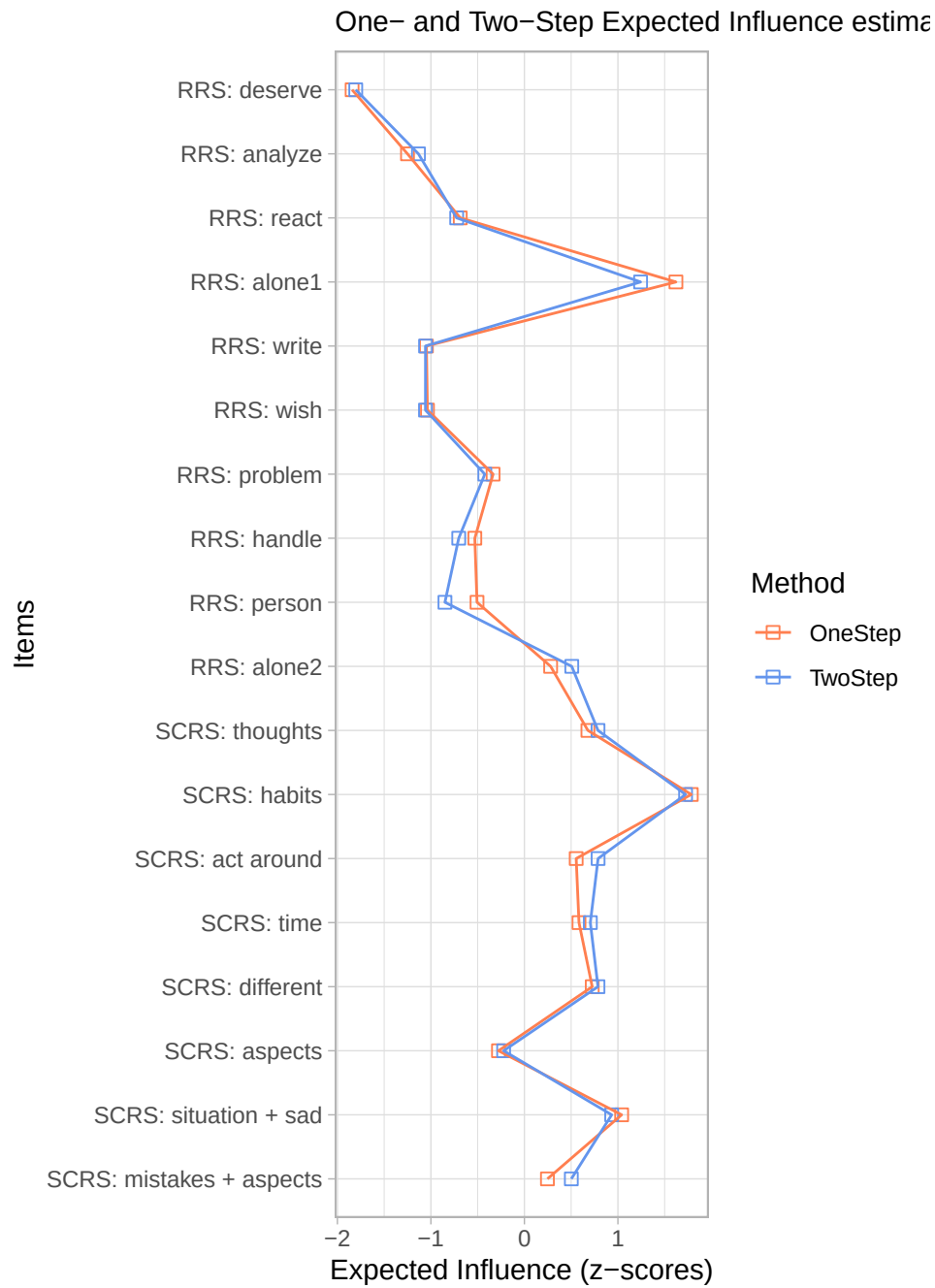


Combined RRS and SCRS Expected Influence Measures

```
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
```



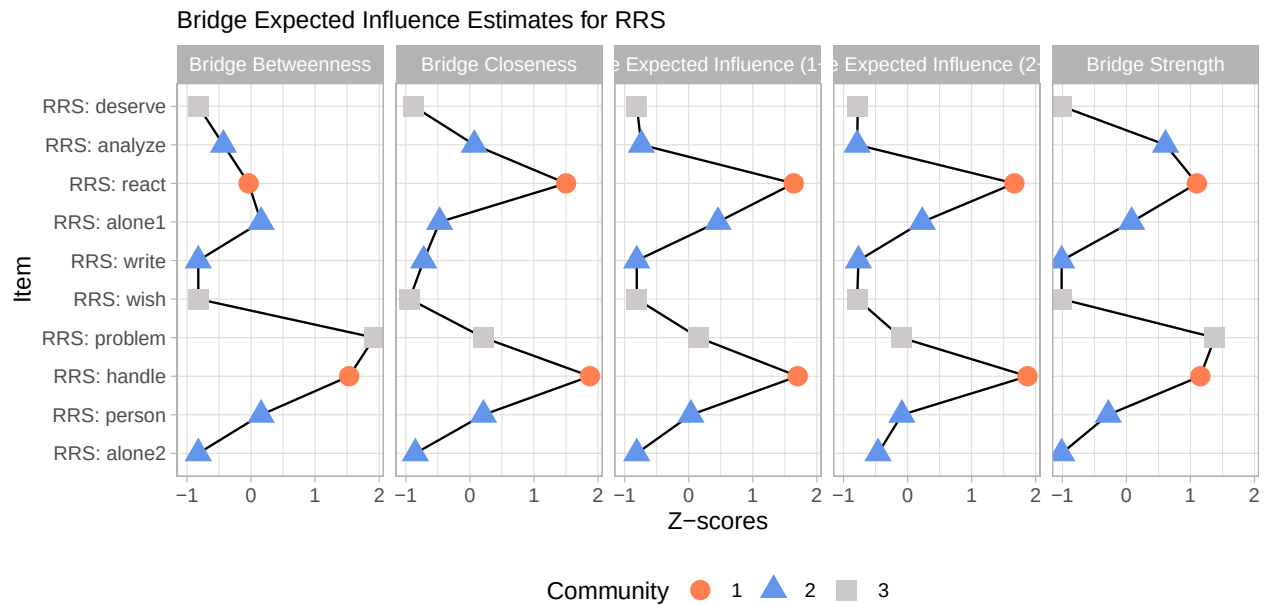
```
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
```



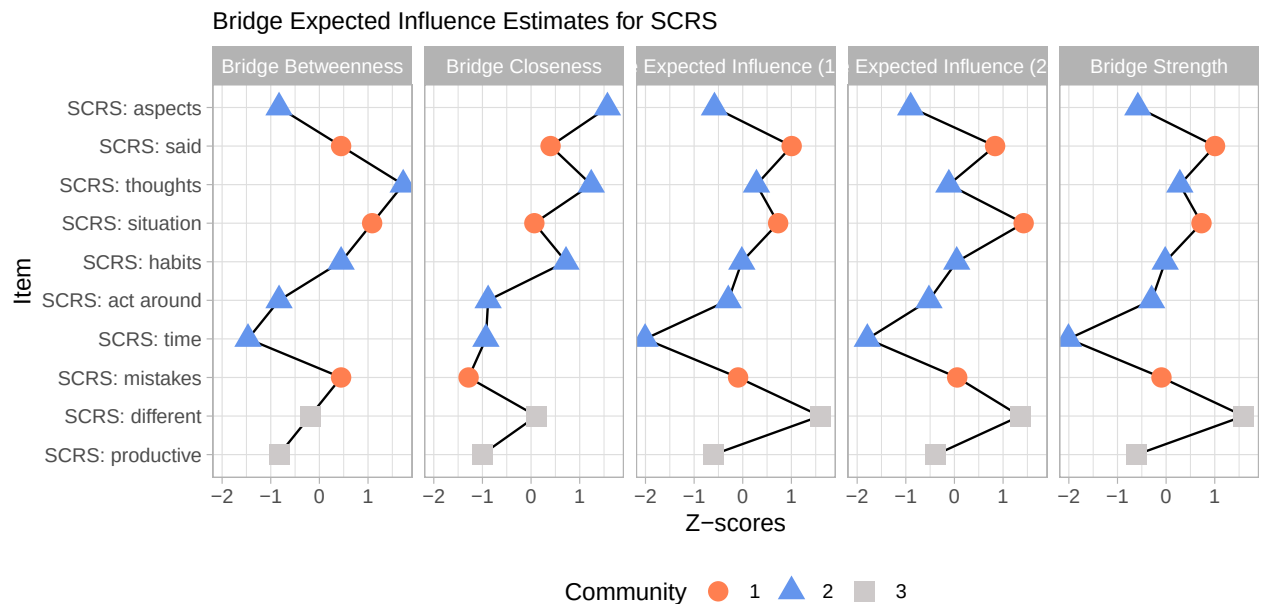
3.4 Bridge nodes

Using the *networktools* R library bridge expected influence measure were estimated for the three tools we used in this study. We also added aesthetics to reflect which items belonged to the same community according to our previous clustering steps.

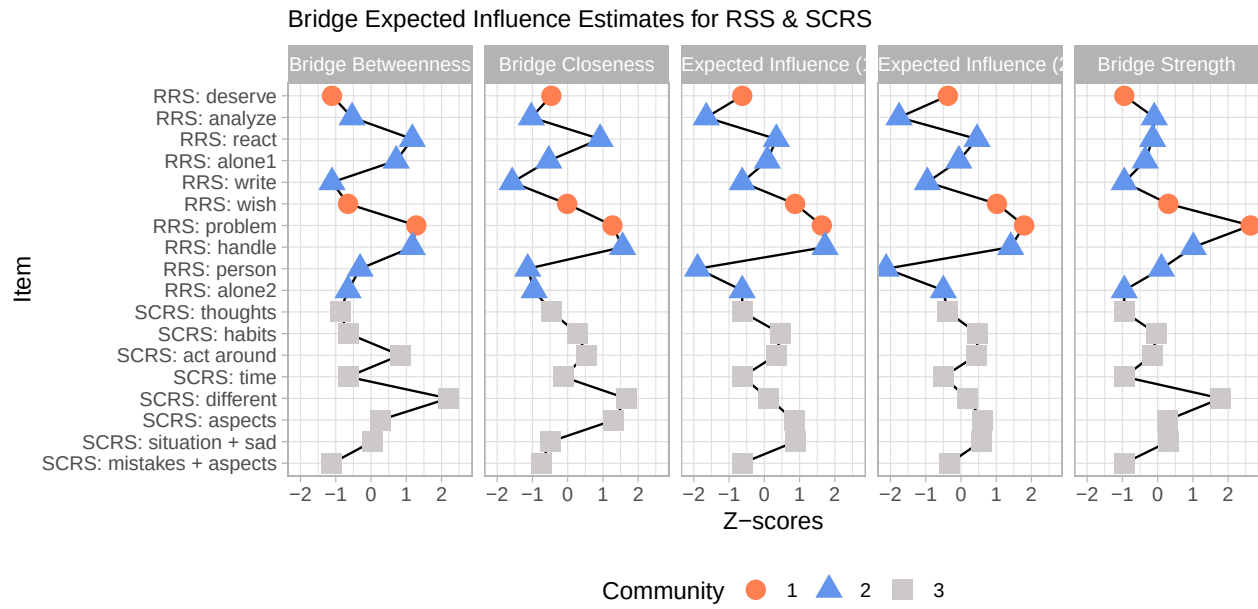
RRS bridge estimates



SCRS bridge estimates



Combined RRS and SCRS bridge estimates



References

- Bernstein, E. E., Heeren, A., & McNally, R. J. (2019). Reexamining trait rumination as a system of repetitive negative thoughts: A network analysis. *Journal of Behavior Therapy and Experimental Psychiatry*, 63, 21–27. <https://doi.org/10.1016/j.jbtep.2018.12.005>
- Epskamp, S., Borsboom, D., & Fried, E. I. (2017). Estimating psychological networks and their accuracy: A tutorial paper. *Behavior Research Methods*, 50(1), 195–212. <https://doi.org/10.3758/s13428-017-0862-1>
- Jones, P. (2017). *Networktools: Assorted and tools for identifying and important nodes and in networks*.
- Newman, M. E. J. (2010). *Networks - an introduction*.